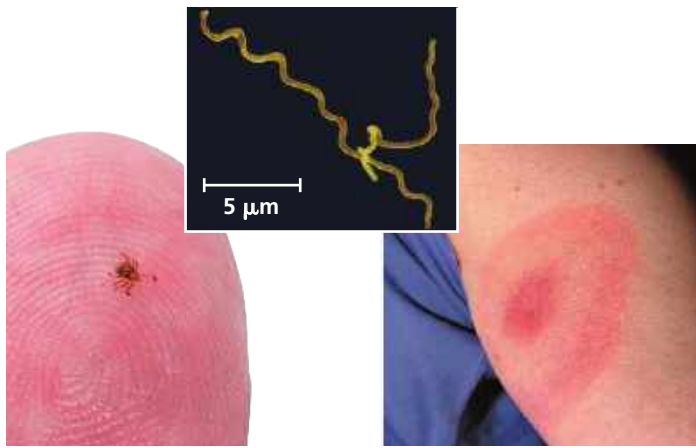




▲ **Figure 27.21 Lyme disease.** Ticks in the genus *Ixodes* spread the disease by transmitting the spirochete *Borrelia burgdorferi* (colorized SEM). A rash may develop at the site of the tick's bite; the rash may be large and ring-shaped (as shown) or much less distinctive.

humans. Signals from the bacterium activate human genes that build the network of intestinal blood vessels necessary to absorb nutrient molecules. Other signals induce human cells to produce antimicrobial compounds to which *B. thetaiotaomicron* is not susceptible. This action may reduce the population sizes of other, competing species, thus potentially benefiting both *B. thetaiotaomicron* and its human host.

Pathogenic Bacteria

All the pathogenic prokaryotes known to date are bacteria, and they deserve their negative reputation. Bacteria cause about half of all human diseases. For example, more than 1.5 million people die each year of the lung disease tuberculosis, caused by *Mycobacterium tuberculosis*. And another 2 million people die each year from diarrheal diseases caused by various bacteria.

Some bacterial diseases are transmitted by other species, such as fleas or ticks. In the United States, the most widespread pest-carried disease is Lyme disease, which infects about 300,000 people each year according to a recent CDC estimate (**Figure 27.21**). Caused by a bacterium carried by ticks, Lyme disease can result in debilitating arthritis, heart disease, nervous disorders, and death if untreated.

Pathogenic prokaryotes usually cause illness by producing poisons, which are classified as exotoxins or endotoxins.

Exotoxins are proteins secreted by certain bacteria and other organisms. Cholera, a dangerous diarrheal disease, is caused by an exotoxin secreted by the proteobacterium *Vibrio cholerae*. The exotoxin stimulates intestinal cells to release chloride ions into the gut, and water follows by osmosis. In another example, the potentially fatal disease botulism is caused by botulinum toxin, an exotoxin secreted by the gram-positive bacterium *Clostridium botulinum* as it ferments various foods, including improperly canned meat,

seafood, and vegetables. Like other exotoxins, the botulinum toxin can produce disease even if the bacteria that manufacture it are no longer present when the food is eaten.

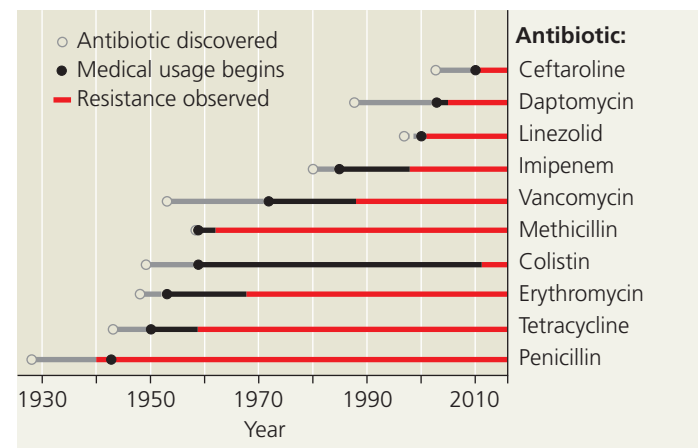
Endotoxins are lipopolysaccharide components of the outer membrane of gram-negative bacteria. In contrast to exotoxins, endotoxins are released only when the bacteria die and their cell walls break down. Endotoxin-producing bacteria include species in the genus *Salmonella*, such as *Salmonella typhi*, which causes typhoid fever. You might have heard of food poisoning caused by other *Salmonella* species that can be found in poultry and some fruits and vegetables.

Finally, horizontal gene transfer can spread genes associated with virulence, turning normally harmless bacteria into potent pathogens. *E. coli*, for instance, is ordinarily a harmless symbiont in the human intestines, but pathogenic strains that cause bloody diarrhea have emerged. One of the most dangerous strains, O157:H7, is a global threat; in the United States alone, there are around 75,000 cases of O157:H7 infection per year, often from contaminated beef or produce. Scientists have discovered over 1,000 genes in O157:H7 that have no counterpart in harmless *E. coli* strains. Some of the genes found only in O157:H7 are associated with virulence and were likely transferred by phage-mediated horizontal gene transfer (transduction) from pathogenic bacterial species.

Antibiotic Resistance

Since their initial use in the 1940s, antibiotics have saved many lives and reduced the incidence of disease caused by pathogenic bacteria. Unfortunately, as **Figure 27.22** shows,

▼ **Figure 27.22 The rise of antibiotic resistance.** As illustrated by the examples shown here, bacteria have developed resistance to every antibiotic currently in clinical use, often within a few years. It took more than 50 years for resistance to develop to colistin, most likely because this antibiotic has toxic side effects in humans and hence was used rarely (as a treatment of last resort).



VISUAL SKILLS Based on this diagram, identify the antibiotic for which resistance appeared before the antibiotic began to be used medicinally.